Case H — Ms. Henderson

19F · Competitive Gymnast · 8-month sharp LBP at L5, extension-provoked | Working Dx: Spondylolysis at L5 (Pars Interarticularis Stress Fracture)

Target: 70%+ | Musculoskeletal (MSK) Physiotherapy

■ PROBLEM LIST

- Painful and restricted lumbar extension
- L5 pars interarticularis stress fracture (structural integrity compromised)
- L5 spinous process tenderness on palpation
- Potential underlying bone density compromise (RED-S risk)
- Paraspinal muscle guarding/spasm secondary to pain
- Unable to perform gymnastics skills requiring extension (walkovers, back handsprings, beam skills)
- Reduced training capacity and skill progression
- Potentially affecting school/social life due to 8-month duration
- Psychological impact: identity as an elite athlete threatened by injury

■ RED FLAGS

- Cauda Equina Syndrome Significance: High-grade spondylolisthesis can compress cauda equina — saddle anaesthesia, bilateral leg weakness, bowel/bladder dysfunction Action: Immediate A&E; referral; urgent MRI
- High-Grade Spondylolisthesis (>Grade II) Significance: >50% anterior slip of L5 on S1 risks L5 nerve root compression and spinal instability; surgical intervention may be required Action: Urgent spinal surgeon/orthopaedic referral; cease sport immediately; MRI + X-ray (lateral standing view)
- Progressive Lower Limb Neurological Deficit Significance: Suggests nerve root or cord compromise from listhesis or associated disc injury Action: Refer to spinal team; cease rehabilitation; urgent imaging

■ PATHOPHYSIOLOGY

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■ DIFFERENTIAL DIAGNOSES

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■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

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■ TREATMENT PLAN

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